

What quality and reliability mean to the consumer

For the user of a product, the product should perform its intended/designed function when used as per the prescribed method and in prescribed operating environment. If it fails to perform a required function or deviates from the required performance, it is declared as a bad-quality product.

Not only should the product perform its function immediately after installation but it should perform without failure over its required operational life time. If it fails very frequently during its operational life-time (for example, a television failing twice or thrice every year), then also it is declared as a bad-quality product.

Quality and reliability engineers relate quality to general performance of a system when it's operated under allowed stresses, and reliability to time-oriented performance without failures.

increased electromagnetic emissions.

4. Use power supply decoupling capacitors—one for every four to five digital integrated circuits (ICs). These are required to counter inductive voltage drop developed by switching currents in digital ICs.

5. Select the track width for signal and power tracks such that these can sink the required current.

6. Provide appropriate spacing between signal tracks to avoid cross-talk.

7. Select pads suitable to component dimensions.

8. Allow adequate clearances between tracks and through-hole component pads so that components can be easily soldered.

9. Use heat-sinks for high-power-consuming components like power amplifiers and power supplies.

10. Perform thermal analysis with EDA tools to find out hotspots on PCBs.

11. Only general guidelines are provided here. Study PCB layout literature for more details.

Packaging

Any electronic product normally has a number of PCBs connected by a motherboard. All the PCBs are placed in a metallic enclosure. The enclosure has connectors fixed to it for external communication of signals. Wiring harness runs from connectors on the chassis to the PCBs/motherboard.

Unit packaging is to be designed carefully such that the unit can withstand thermal, mechanical, electromagnetic and electrostatic discharge (ESD) stresses.

Thermal stress. Performance of

many electronic components and systems degrades with temperature. This happens more when subjected to high temperatures for a long duration. Electronic components can also get mechanically damaged when temperature changes fast from high to low and vice versa.

The most common method of thermal control is use of heat-sinks. By employing a heat-sink, a low thermal resistance path is provided from the component to the air. Heat conducts from the component to the heat sink. Heat sink is cooled by convection, with air serving as the thermal reservoir.

Other techniques are:

1. In a forced-air-cooled package, spread the heat-dissipating parts uniformly along the cold wall.

2. Do not place thermally sensitive or highly dissipating parts close to each other.

3. Do not place thermally sensitive parts next to hotspots.

4. In free-convection-cooled equipment, do not place parts directly above highly dissipating parts; stagger them horizontally.

5. For contact interfaces, use as much contact area as possible.

Mechanical stress. Excessive mechanical stresses, such as vibration and shock, generated during operation of electronic sub-systems (for example, launch of missile systems) can cause chafed wiring, loose fasteners or components, intermittent electrical contacts, deformed seals, failed components, and cracked and/or broken structures.

Protection against mechanical stresses is generally achieved by suit-

able packaging, mounting and structural techniques. Two approaches used are mechanically isolating the equipment and building the equipment to required strength.

Packaging analysis using CAD tools is done to determine natural frequencies and mechanical stresses within components, produced by shock and vibrations. If stresses exceed safe levels, corrective measures such as stiffening and incorporation of further support members are required. An isolation system can be used at the source of vibration in addition to isolating the protected component. Damping devices can be used to reduce peak oscillations.

Humidity. Many electronic systems need to be stored and operated in coastal areas, which are highly humid. Humidity can cause corrosion, loss of mechanical strength, change of electrical properties and increased conductivity of insulators, thus degrading the performance of electronic systems.

Protection against moisture can be provided by applying protective coating, using hermetically sealed components and encapsulating materials with moisture-resistant waxes.

Electromagnetic interference (EMI). Electronic systems' operation could be affected by electromagnetic interference caused by electromagnetic wave radiations. Sources of electromagnetic wave radiation include communication transmitters, radio broadcasting transmitters or radiation from circuits operating within a unit itself. Examples of EMI problems include noise created on a television set when a nearby grinder is operating, intermittent resetting of computers and noise heard in a radio set when a fluorescent lamp is switched on.

EMI problem also occurs when an electronic unit shares its source of power with other electronic units. This situation is called conducted susceptibility.

Effects of EMI can be minimised by:

1. Providing a metallic enclosure as equipment casing that isolates the circuits within from electromagnetic radiation